ENGINEERING BACKLOG & QUALITY GATES

EduMore Notebook

Implementation tasks, acceptance tests and failure-handling requirements

22	P0–P2	TESTED
IMPLEMENTATION TASKS	PRIORITY LEVELS	QUALITY FIRST

Planning baseline • Version 1.1 • 20 October 2026

How to Use This Notebook

This notebook converts the product and architecture decisions into a practical implementation backlog. Each task has a concise English description, acceptance criteria, non-acceptance criteria, and error-handling expectations. Priorities and phases are planning recommendations, not delivery estimates.

Priority key: P0 = release-critical • P1 = high priority • P2 = later or conditional

Release roadmap

PHASE	GOAL	EXIT GATE
01 Foundation	Auth, tenant isolation, data model, task center, recovery, security, observability, versioning.	Core security, migration, recovery and job tests pass.
02 First workflow	Research → notes → diagrams → quality checks → faculty approval → PDF.	End-to-end staging tests pass; approved PDF only; failure paths tested.
03 AI workflows	Curriculum Designer and assessment/test generation.	Source traceability, validation, review and export tests pass.
04 Academic management	Progress, workload, scheduling, alerts and reports.	Metric definitions, data freshness, permissions and approvals tested.

Definition of Done

- Automated tests cover happy paths, validation failures, permissions, and recovery paths.
- Tenant isolation is tested at API, search, export, and file-access layers.
- Unapproved notes, tests, and official reports cannot be published as final.
- Long-running jobs have persisted status, bounded retries, and safe recovery.
- Errors are actionable, observable, and do not leak secrets or unnecessary sensitive data.
- Documentation, migration/recovery notes, monitoring, and rollback steps are updated.

Implementation Task Backlog

Tasks are grouped by recommended delivery phase. A task is not complete until its acceptance criteria pass and its rejection conditions are demonstrably prevented.

EM-001 Institution and Role Foundation

P0 • Phase 1 — Platform Foundation

Description Create institution-aware user accounts and role-based access for Admin, Management, and Faculty.

ACCEPTANCE CRITERIA

• Users can sign in and sign out securely.

• Each user is associated with an institution and an assigned role.

• Every protected API checks authorization server-side.

• Automated tests prove users cannot access another institution's records.

NON-ACCEPTANCE CRITERIA

• A user can access data by changing an ID in the URL or request body.

• Permissions are enforced only in the frontend.

• Passwords or secrets are stored in plaintext.

ERROR HANDLING

• Invalid credentials return a generic authentication error.

• Expired/invalid sessions are rejected and can be renewed or re-authenticated safely.

• Unexpected auth failures are logged without passwords, tokens, or sensitive payloads.

• Tenant context missing or invalid fails closed.

EM-002 Course and Curriculum Data Model

P0 • Phase 1 — Platform Foundation

Description Define institution-scoped records for courses, curriculum versions, chapters, learning outcomes, and source references.

ACCEPTANCE CRITERIA

• Core entities have stable IDs, timestamps, institution ownership, and validation rules.

• Curriculum versions and chapter ordering are represented explicitly.

• Foreign keys and indexes protect relationships and common queries.

NON-ACCEPTANCE CRITERIA

• Resources are stored as unstructured blobs with no traceable relationships.

• Cross-institution relationships can be created accidentally.

• Published curriculum is overwritten without a new version.

ERROR HANDLING

• Invalid relationships are rejected with actionable validation errors.

• Database failures roll back partial transactions.

• Migrations are versioned, reviewed, and tested against a representative database.

EM-003 Background Job and Task Center

P0 • Phase 1 — Platform Foundation

Description Implement background job submission, persisted status, progress reporting, retry controls, and user-visible task history.

ACCEPTANCE CRITERIA

• Long-running jobs expose queued, running, review-required, completed, and failed states as applicable.

• The frontend can refresh and retrieve current status after reload.

• Retries are bounded and authorized; repeated requests do not create duplicate side effects.

• Failures include a safe user message and a trace/correlation ID.

NON-ACCEPTANCE CRITERIA

• The user must keep a browser tab open for the job to finish.

• A failed task is shown as completed.

• Automatic retries run indefinitely or create duplicate documents.

ERROR HANDLING

• Worker restarts resume or safely requeue eligible work.

• Transient failures use bounded backoff; permanent failures are surfaced for action.

• Queue/database outage is monitored and new work is rejected or queued safely rather than falsely accepted.

EM-004 Research and Source Verification Service

P0 • Phase 2 — First End-to-End Workflow

Description Search for relevant learning sources, prioritize credible sources, verify claims across sources, and retain citations.

ACCEPTANCE CRITERIA

• Research results include source title, URL, retrieval time, and relevant evidence/snippets where permitted.

• Official documentation and reputable sources receive priority based on task context.

• Conflicting or weak evidence is flagged; generated content does not invent citations.

• Source links are attached to the generated resource for review.

NON-ACCEPTANCE CRITERIA

• A URL is presented as evidence without being retrieved or checked.

• A generated claim is cited to an unrelated source.

• The workflow copies long source passages instead of synthesizing original explanations.

ERROR HANDLING

• Search provider timeouts use bounded retries and fallback providers where configured.

• If research is incomplete, the result is marked partial and routed to review rather than described as verified.

• Unreachable sources are labeled unavailable; cached evidence is marked with its retrieval date.

EM-005 Notes Generation Workflow

P0 • Phase 2 — First End-to-End Workflow

Description Generate structured, original notes from a topic or approved course chapter using researched evidence and learning-level requirements.

ACCEPTANCE CRITERIA

• Notes include a clear structure, definitions, examples, and learning-level-appropriate explanations.

• Relevant source references are retained.

• Coverage and required-section checks run before review.

• Content status remains draft until approval.

NON-ACCEPTANCE CRITERIA

• Notes are published automatically without required approval.

• The workflow invents unsupported facts or citations without flagging uncertainty.

• Missing required sections are silently ignored.

ERROR HANDLING

• Insufficient sources or contradictory evidence produce a review flag.

• Model timeout or malformed output triggers bounded retry or safe failure.

• Partial output is saved as a clearly labeled draft only if it passes basic schema/safety checks.

EM-006 Technical Diagram and Illustration Service

P1 • Phase 2 — First End-to-End Workflow

Description Generate code-based technical diagrams and use AI-generated illustrations selectively, with validation before inclusion.

ACCEPTANCE CRITERIA

• Technical diagrams use reproducible formats such as Mermaid/SVG/charts where suitable.

• Diagram labels, references, and rendering are checked.

• The PDF preview is inspected for clipping, unreadable text, or broken assets.

NON-ACCEPTANCE CRITERIA

• Broken or empty diagrams are included in final PDFs.

• A generated illustration is treated as a precise technical diagram without validation.

• External image URLs are embedded without safe retrieval and licensing checks.

ERROR HANDLING

• Diagram syntax errors trigger a repair attempt within a limit, then a text fallback or review flag.

• Image generation/provider failure does not block the whole notes task when diagrams are optional.

• Rendering failure prevents final publication and records diagnostic details.

EM-007 Multi-Layer Content Quality Review

P0 • Phase 2 — First End-to-End Workflow

Description Combine deterministic validation, AI review, evidence checks, and bounded revision attempts for generated educational content.

ACCEPTANCE CRITERIA

- Rules check required sections, formatting, learning outcomes, and schema validity.
- AI review checks clarity, completeness, suitability, and potential factual issues.
- Evidence checks flag claims not supported by available sources.
- Retry limits are enforced and unresolved issues are sent to faculty review.

NON-ACCEPTANCE CRITERIA

- An AI reviewer is treated as a guarantee of factual correctness.
- The workflow retries forever.
- A failed quality check is hidden from the faculty reviewer.

ERROR HANDLING

- Reviewer disagreement is recorded and flagged for human review.
- Unavailable review models trigger a controlled fallback or a blocked/partial status.
- Validation exceptions fail closed for publication.

EM-008 Faculty Review and Approval

P0 • Phase 2 — First End-to-End Workflow

Description Provide a review interface for notes, diagrams, citations, and quality flags before final publication.

ACCEPTANCE CRITERIA

- Faculty can preview, edit, approve, reject, or request revisions based on permissions.
- Approval records include actor, timestamp, resource version, and decision.
- Only the approved version can be published or exported as final.
- A revision creates a new version or draft without erasing approved history.

NON-ACCEPTANCE CRITERIA

- A non-authorized user can approve content.
- A draft is labeled final before approval.
- Approval history can be silently altered.

ERROR HANDLING

- Concurrent edits are detected and resolved without silent overwrite.
- Approval API failure leaves the resource in a non-published state.
- Notifications that fail can be retried without duplicating approval actions.

EM-009 PDF Rendering and File Storage

P0 • Phase 2 — First End-to-End Workflow

Description Create professionally formatted PDFs from approved content and store them in object storage with secure access.

ACCEPTANCE CRITERIA

- PDF includes headings, page numbers, readable typography, diagrams, source references, and consistent spacing.
- A render/preview check detects missing content, clipped elements, or broken assets.
- Files are associated with the correct institution and resource version.
- Download access is authorized and can use expiring links.

NON-ACCEPTANCE CRITERIA

- An unapproved draft is presented as the final PDF.
- PDF links expose files across institutions.
- A failed upload is recorded as a successful export.

ERROR HANDLING

- Renderer failures trigger a bounded retry and preserve the approved source content.
- Object storage failures leave export status failed/pending rather than completed.
- Temporary download URLs expire; missing files produce a safe error and an operational alert.

EM-010 End-to-End Notes-to-PDF Orchestration

P0 • Phase 2 — First End-to-End Workflow

Description Connect topic input, research, notes generation, diagrams, quality checks, faculty approval, and PDF export in one recoverable workflow.

ACCEPTANCE CRITERIA

- Workflow stages and state are persisted with checkpoints.
- The task center shows progress and current required action.
- A failed stage resumes from a safe checkpoint where possible.
- The final PDF is produced only for the approved version.

NON-ACCEPTANCE CRITERIA

- A later failure erases the source input or approved content.
- The workflow claims success while a required stage failed.
- Retries create duplicate final documents or duplicate approval events.

ERROR HANDLING

- Each stage has timeouts, bounded retries, structured errors, and correlation IDs.
- Non-recoverable failures preserve prior safe state and provide a retry/review path.
- Provider fallback outputs pass the same quality gates.

EM-011 Curriculum Designer Workflow

P1 • Phase 3 — Remaining AI Workflows

Description Research a course goal, generate a sequenced curriculum and visual roadmap, validate it, and request faculty approval.

ACCEPTANCE CRITERIA

- Input captures course, learner level, target duration, goals, and relevant constraints.
- Curriculum includes modules, learning outcomes, sequence, estimated time, and sources.
- Roadmap data is structured and visually rendered.
- Faculty can review and approve a specific version.

NON-ACCEPTANCE CRITERIA

- Time estimates contradict the requested duration without a warning.
- Curriculum sources are fabricated or not linked.
- The system publishes without the required review.

ERROR HANDLING

- Missing requirements trigger a clarification or clearly stated assumptions.
- Conflicting sources or impossible timelines are flagged.
- Roadmap rendering failure does not destroy the curriculum text; a text outline is available.

EM-012 Assessment and Test Generator

P1 • Phase 3 — Remaining AI Workflows

Description Generate tests linked to course chapters with difficulty, topic coverage, marks, time limits, answers, explanations, and export options.

ACCEPTANCE CRITERIA

- Questions map to defined topics/learning outcomes and requested difficulty levels.
- Answer keys and explanations are validated where feasible.
- Duplicate and coverage checks run.
- Faculty approval is required before official publication; approved tests can be exported to PDF.

NON-ACCEPTANCE CRITERIA

- A test contains missing answers when answers are required.
- Marks/time limits are inconsistent or total marks do not reconcile.
- The test is published before approval.

ERROR HANDLING

- Ambiguous questions are flagged for revision.
- Answer-validation disagreement is surfaced for faculty review.
- PDF or model failure leaves the test draft intact and does not mark the task complete.

EM-013 Academic Progress and Workload Tracking

P1 • Phase 4 — Academic Management

Description Track course/syllabus completion, pending topics, faculty workload, and planned versus actual academic progress.

ACCEPTANCE CRITERIA

• Progress is derived from explicitly defined and traceable records.

• Users only see data allowed by their institution and role.

• Reports show data freshness and the basis for calculations.

• Manual corrections, if supported, are audited.

NON-ACCEPTANCE CRITERIA

• Progress metrics are fabricated or silently inferred from missing data.

• A report hides stale or incomplete data.

• Faculty can edit another institution's workload.

ERROR HANDLING

• Missing source records produce an incomplete-data state, not a fabricated zero.

• Calculation failures are logged and the last valid report is labeled with its timestamp.

• Concurrent updates are handled transactionally.

EM-014 Scheduling and Delay Alerts

P2 • Phase 4 — Academic Management

Description Support course planning, timetable/scheduling, pending-work visibility, and configurable delay alerts.

ACCEPTANCE CRITERIA

• Schedules respect configured dates, durations, and constraints.

• Alerts include the reason, relevant deadline, and actionable next step.

• Users can distinguish planned, delayed, and completed items.

• Scheduling changes are audited.

NON-ACCEPTANCE CRITERIA

• Alerts are generated from unverified or missing dates without qualification.

• The system silently overwrites conflicting schedules.

• A notification is marked delivered without confirmation from the delivery service.

ERROR HANDLING

• Scheduling conflicts are surfaced for resolution.

• Notification failures are retried safely and remain visible.

• Timezone/date parsing errors are validated before saving.

EM-015 Academic Report Generation

P1 • Phase 4 — Academic Management

Description Create management reports from verified academic progress, workload, and scheduling data with traceable metrics.

ACCEPTANCE CRITERIA

• Each metric has a defined calculation and source period.

• Reports show generation time and data completeness.

• Official reports require approval before publication where configured.

• Exports preserve readable layout and access controls.

NON-ACCEPTANCE CRITERIA

• The report invents metrics when data is missing.

• An official report is published without required approval.

• Sensitive institution data is exposed to unauthorized users.

ERROR HANDLING

• Data query failure produces a failed or partial report with clear labeling.

• Stale data is flagged.

• Export failure does not destroy the report draft or source calculation.

EM-016 Model Routing and Provider Fallback

P1 • Phase 1 — Platform Foundation

Description Provide configurable model routing by task complexity, cost, latency, and quality needs, with controlled provider fallback.

ACCEPTANCE CRITERIA

• Provider/model settings are externalized from business logic.

• Fallback occurs only for configured transient failures or availability conditions.

• Outputs from fallback models go through the same validation gates.

• Usage, latency, and cost are recorded without logging sensitive prompts unnecessarily.

NON-ACCEPTANCE CRITERIA

• A provider fallback bypasses content validation.

• Secrets are hardcoded in source control.

• Provider errors trigger uncontrolled repeated calls.

ERROR HANDLING

• Rate limits use bounded backoff and respect provider retry guidance.

• If all providers fail, the task remains failed/paused with safe state and a clear message.

• Configuration errors fail early and are observable.

EM-017 Checkpointing, Idempotency, and Recovery

P0 • Phase 1 — Platform Foundation

Description Persist workflow checkpoints and make retryable actions safe to repeat without duplicate records, approvals, or files.

ACCEPTANCE CRITERIA

• Checkpoint schema and workflow versions are tracked.

• Safe workflows resume after worker restart.

• Idempotency keys or equivalent controls protect side effects.

• Recovery paths are covered by fault-injection or restart tests.

NON-ACCEPTANCE CRITERIA

• A retry silently duplicates published resources.

• Old checkpoints are resumed against incompatible workflow versions without migration/validation.

• Sensitive data is stored in checkpoints without need or controls.

ERROR HANDLING

• Incompatible checkpoints stop safely and offer a controlled restart/recovery path.

• Checkpoint-store failure prevents claiming recoverable progress.

• Recovery failures generate alerts and preserve the last known safe state.

EM-018 Resource Versioning and Change Impact

P1 • Phase 1 — Platform Foundation

Description Version curricula, notes, tests, and reports and identify resources potentially affected by a curriculum change.

ACCEPTANCE CRITERIA

• Each published version is immutable or has an auditable revision history.

• Resources link to course, chapter, and curriculum version.

• Curriculum changes flag potentially outdated downstream resources.

• Faculty can compare and review versions.

NON-ACCEPTANCE CRITERIA

• Approved resources are silently overwritten.

• A curriculum change claims all downstream content is updated without validation.

• Version history can be deleted by ordinary faculty actions.

ERROR HANDLING

• Version conflicts are detected and surfaced.

• Failed migrations preserve existing records and support rollback.

• Impact analysis uncertainty is labeled for review.

EM-019 Observability and Audit Trail

P1 • Phase 1 — Platform Foundation

Description Instrument APIs and workflows and maintain privacy-safe, institution-aware audit events for access, approvals, and changes.

ACCEPTANCE CRITERIA

- Logs and metrics include correlation IDs, status, latency, and failure category.
- AI usage and costs are measurable per workflow and institution where permitted.
- Audit events record actor, action, resource/version, and timestamp.
- Secrets, credentials, and unnecessary sensitive content are excluded from logs.

NON-ACCEPTANCE CRITERIA

- Logs expose passwords, tokens, or unnecessary personal data.
- Critical workflow failures are invisible to operators.
- Audit records can be silently changed by ordinary users.

ERROR HANDLING

- Monitoring-provider outage does not crash core requests; local/application logging remains available where designed.
- Alert delivery failures are retried or surfaced.
- Log retention and access follow documented policy.

EM-020 Role-Based Dashboards and Task Center

P1 • Phase 1 — Platform Foundation

Description Provide separate Faculty and Management dashboards with task status, approvals, progress, alerts, and permitted next actions.

ACCEPTANCE CRITERIA

- Dashboard data is filtered by institution and role on the server.
- Tasks show current state, timestamps, and actionable failure/review details.
- Users can resume, retry, review, approve, or download only when authorized.
- Empty, loading, and error states are clear and accessible.

NON-ACCEPTANCE CRITERIA

- Dashboard metrics are treated as authoritative when stale without a freshness indicator.
- Unauthorized actions appear functional or succeed through direct API calls.
- A loading indicator remains forever after a failed request.

ERROR HANDLING

- API errors show a retry path and correlation ID where appropriate.
- Polling/backoff is bounded and does not overload the server.
- A stale dashboard is labeled and refreshable.

EM-021 Security and Tenant Isolation Tests

P0 • Phase 1 — Platform Foundation

Description Implement layered security controls and automated tests for cross-tenant isolation, permissions, file access, and common abuse cases.

ACCEPTANCE CRITERIA

• Cross-tenant access tests cover API reads/writes, search, exports, and file downloads.

• Rate limits and input validation are applied to appropriate endpoints.

• Secrets are managed outside source control; encryption is configured for supported paths.

• Security-sensitive changes require review and regression tests.

NON-ACCEPTANCE CRITERIA

• Tenant isolation relies only on frontend filtering.

• Public or guessable file URLs expose protected documents.

• Security tests are skipped for convenience without a documented risk decision.

ERROR HANDLING

• Authorization failures deny access by default.

• Security scanning/test infrastructure failures are reported and block release when required by policy.

• Suspected abuse is logged with privacy-safe details and can trigger throttling.

EM-022 Phased Release and Quality Gates

P0 • All phases

Description Release the platform in small, tested phases, starting with the notes-to-approved-PDF workflow and adding other workflows only after quality gates pass.

ACCEPTANCE CRITERIA

• Each phase has documented scope, tests, known risks, rollback steps, and release approval.

• Critical acceptance tests and tenant-isolation tests pass before release.

• The first workflow works end-to-end in a staging environment using realistic test data.

• Known limitations are documented; monitoring and rollback are available.

NON-ACCEPTANCE CRITERIA

• All modules are launched before end-to-end integration is tested.

• Release depends only on a successful demo or manual happy-path run.

• Known critical data-loss or cross-tenant defects are accepted without an explicit risk decision.

ERROR HANDLING

• Deployment failure triggers rollback or safe maintenance mode.

• Database migrations are backward-compatible where practical and have a recovery plan.

• Post-release error spikes trigger investigation and, if needed, rollback.

Cross-Cutting Error Handling Standard

Apply these controls to every API and workflow. Individual tasks may require stricter handling.

Input validation

Validate request shape, length, file type, size, and business constraints at the API boundary. Return field-specific errors without echoing secrets.

Timeouts and retries

Set timeouts for external calls. Retry only transient failures, use bounded exponential backoff with jitter, and respect provider retry guidance.

Idempotency

Use idempotency keys or equivalent deduplication for job submission, approvals, export creation, notifications, and other side effects.

Partial failure

Represent partial/incomplete results explicitly. Do not label them verified, approved, or complete. Preserve safe drafts when useful.

Checkpoint recovery

Persist checkpoints at safe boundaries. Validate checkpoint/workflow version before resume. If resume is unsafe, stop and offer a controlled restart.

Provider outage

Use configured fallback providers only where permitted. Apply the same validation gates to fallback output. If all providers fail, preserve state and report failure.

Storage failure

Do not mark uploads/exports complete until storage confirms success. Reconcile orphaned objects and records with a safe cleanup/recovery process.

Concurrency

Use transactions, version checks, or optimistic locking to prevent silent overwrite and conflicting approvals.

Security

Fail closed when tenant or permission context is missing. Never expose stack traces, access tokens, passwords, private prompts, or other secrets to users.

Observability

Attach a correlation ID to each task. Record stage, duration, outcome, and safe error category; alert on critical failure patterns.

User communication

Show status, a useful next step, and whether retry is safe. Never claim a task succeeded until all required stages are confirmed.

Disaster recovery

Back up structured data and important configuration, test restore procedures, define retention, and document recovery objectives before production.

Open Technical Decisions

- Choose hosting, deployment, database and object-storage providers.
- Choose background queue/worker and checkpoint persistence technology.
- Select model providers and define evaluation sets, cost limits and fallback rules.
- Define availability, latency, backup/restore, retention and incident-response targets.
- Confirm privacy, licensing, and institution-specific compliance requirements.
- Decision 20: Notes & PDF Generator was recommended as the first workflow but was not explicitly confirmed in the role-play.

Important Reliability Note

The system cannot be guaranteed never to fail. The engineering goal is graceful degradation, safe retries, checkpoint recovery, data integrity, observability, and clear communication. Never mark a workflow complete until all required stages have succeeded.

Document status: planning artifact, not proof of implementation or production testing.